

Matplotlib Parameters Quick Reference

3, 4 & 5. plt.xlabel() / plt.ylabel() / plt.title()

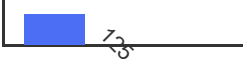
Unified text elements sharing identical core styling properties (use `labelpad` for axes and `pad` for title).

PARAMETER	MEANING	EXAMPLE	VISUAL
label / string	Text content	<code>plt.xlabel("Airline")</code> <code>plt.ylabel("Price")</code> <code>plt.title("Title")</code>	Text Content →
fontsize	Text size	<code>fontsize=14</code> (or <code>16</code>)	BIG TEXT
fontweight	Text thickness	<code>fontweight='bold'</code>	Bold Text
color	Text color	<code>color='red'</code>	● Colored Text
labelpad / pad	Distance from axis/chart	<code>labelpad=10</code> or <code>pad=20</code>	Text ↑ Element
loc	Position/Alignment	<code>loc='center'</code> / <code>'left'</code> / <code>'right'</code>	← Aligned →

1. plt.xticks()

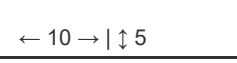
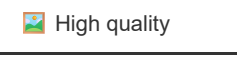
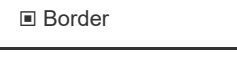
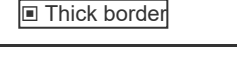
PARAMETER	MEANING	EXAMPLE	VISUAL
ticks	Position of ticks	<code>plt.xticks([0,1,2])</code>	0 1 2
labels	Custom tick names	<code>plt.xticks([0,1], ['A','B'])</code>	A B
rotation	Rotate labels	<code>plt.xticks(rotation=45)</code>	↗
fontsize	Label size	<code>plt.xticks(fontsize=12)</code>	A
fontweight	Label thickness	<code>plt.xticks(fontweight='bold')</code>	Air India
color	Label color	<code>plt.xticks(color='red')</code>	● Air India
ha	Horizontal alignment	<code>plt.xticks(ha='right')</code>	Air India →
va	Vertical alignment	<code>plt.xticks(va='top')</code>	Air India ↓

2. plt.bar_label()

PARAMETER	MEANING	EXAMPLE	VISUAL
container	Bars to label	<code>ax.containers[0]</code>	
labels	Custom labels	<code>labels=['A','B']</code>	
fmt	Number format	<code>fmt='%d'</code>	
fmt='%f'	2 decimal places	<code>fmt='%f'</code>	
padding	Space from bar	<code>padding=5</code>	
rotation	Rotate labels	<code>rotation=45</code>	
fontsize	Label size	<code>fontsize=10</code>	
color	Label color	<code>color='red'</code>	
label_type='edge'	Label at bar edge	<code>label_type='edge'</code>	
label_type='center'	Label in center	<code>label_type='center'</code>	

6. plt.figure()

Controls overall chart/figure (e.g., `plt.figure(figsize=(10, 5), dpi=150)`)

PARAMETER	MEANING	EXAMPLE	VISUAL
figsize	Width & height	<code>figsize=(10,5)</code>	
dpi	Image resolution	<code>dpi=150</code>	
facecolor	Figure background	<code>facecolor='white'</code>	
edgecolor	Figure border	<code>edgecolor='black'</code>	
linewidth	Border thickness	<code>linewidth=2</code>	

8. plt.ylim()

Controls Y-axis range (e.g., `plt.ylim(0, 50000)`)

PARAMETER	MEANING	EXAMPLE	VISUAL
bottom	Minimum Y value	<code>plt.ylim(bottom=0)</code>	0 —————
top	Maximum Y value	<code>plt.ylim(top=50000)</code>	————— 50K
Both	Set range	<code>plt.ylim(0, 50000)</code>	0 — 50K

9. plt.xlim()

Controls X-axis range.

PARAMETER	MEANING	EXAMPLE	VISUAL
left	Minimum X	<code>plt.xlim(left=0)</code>	0 —————→
right	Maximum X	<code>plt.xlim(right=10)</code>	—————→ 10
Both	Set range	<code>plt.xlim(0,10)</code>	0 — 10

10. plt.grid()

Adds grid lines (e.g., `plt.grid(axis='y', alpha=0.3)`)

PARAMETER	MEANING	EXAMPLE	VISUAL
axis	Which grid	<code>axis='y'</code>	Horizontal lines
alpha	Transparency	<code>alpha=0.5</code>	Light grid
linestyle	Line style	<code>linestyle='--'</code>	----
linewidth	Line thickness	<code>linewidth=0.5</code>	Thin lines
color	Grid color	<code>color='gray'</code>	Gray grid
visible	Show/hide	<code>visible=True</code>	Grid ON

11. plt.legend()

Controls chart legend.

PARAMETER	MEANING	EXAMPLE	VISUAL
loc	Legend position	<code>loc='upper right'</code>	↗ Legend

PARAMETER	MEANING	EXAMPLE	VISUAL
fontsize	Text size	fontsize=10	Price
title	Legend heading	title='Category'	Category Price
ncol	Number of columns	ncol=2	A B
frameon	Show border	frameon=True	Legend